

2.1.2 Stepwise Reasoning

مرحلہ وار استدلال میں الگورتھم کو منطقی حصوں میں توڑ کر ہر مرحلے پر اس کی حالت (State) دیکھی جاتی ہے: کیا شروع...

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 03 of 24

2.1.2 Stepwise Reasoning

LEARN IT

Stepwise reasoning (also called **logic verification**) breaks an algorithm into logical sections and checks the **state** of the algorithm at each stage. A **state** is a snapshot of what is true about the variables at that moment (for example, "Sum currently holds the total of the numbers seen so far").

You do **not** fill a row for every line. You think through the logic:

1. Analyse each step of the algorithm.
2. Check: does it **start in a valid state**?
3. Does it **handle all possible cases**?
4. Are the **conditions correct**?
5. If the loop is correct at the **start** of an iteration, does it remain correct at the **end**?

WORKED EXAMPLE

Worked example: sum of the numbers from 1 to N.

Step 1: Start
 Step 2: Input N
 Step 3: SET Sum = 0
 Step 4: For i = 1 to N
 Step 5: Sum = Sum + i
 Step 6: END For
 Step 7: Print Sum
 Step 8: End

Stepwise reasoning breakdown

Check	What you verify
Initial state	Sum = 0 before the loop. Valid: nothing has been added yet.
Loop boundary	The loop runs exactly from i = 1 to i = N. No missing number, no extra number.
Iteration	Each i is added once: after k iterations, Sum = 1+2+...+k.
Final state	When the loop ends, Sum holds 1+2+...+N, which is the required total.
Edge case N = 0	The loop body never runs, Sum stays 0. Correct (the sum of no numbers is 0).
Check case N = 5	Sum becomes 15, which matches $5 \times 6 / 2 = 15$.

```
N = 5
Sum = 0
for i in range(1, N + 1):
    Sum = Sum + i
print(Sum)
```

OUTPUT

15

```
N = 0
Sum = 0
for i in range(1, N + 1):
    Sum = Sum + i
print(Sum)
```

OUTPUT

0

REMEMBER IT

Think of stepwise reasoning as a **teacher checking a proof**, not as a cashier counting coins one by one. You ask "is this block always true?" instead of "what is Sum when $i = 17$?"

COMMON MISTAKE

Mistake: rewriting the whole trace table and calling it stepwise reasoning.

Fix: write short comments about **state** (start, loop, finish, edges). If you listed every i from 1 to 1000, you used the wrong method.

اردو میں سمجھیں

مرحلہ وار استدلال میں الگورتھم کو منطقی حصوں میں توڑ کر ہر مرحلے پر اس کی حالت (State) دیکھی جاتی ہے: کیا شروع درست ہے، کیا تمام صورتیں سنبھلی ہیں، کیا شرط صحیح ہے، اور کیا لوپ کے ہر چکر کے بعد بھی نتیجہ درست رہتا ہے۔ ایک سے N تک جمع کی مثال میں Sum شروع میں 0 ہے، لوپ 1 سے N تک چلتا ہے، اور $N=0$ پر جواب 0 ہی رہتا ہے۔

CHECK YOURSELF

1. What other name is given to stepwise reasoning?
2. For the sum algorithm, why is $\text{Sum} = 0$ a valid start?
3. What happens when $N = 0$, and is that correct?

Answers: 1. Logic verification. 2. Because no numbers have been added yet, so the running total should be zero. 3. The loop does not run and Sum remains 0, which is the correct sum of no numbers.

Next lecture: 2.1.3 Comparing Trace Tables and Stepwise Reasoning